

Three-step equations



1 Solve the following equations:

a $3(4s + 1) = -21$

c $2(3t - 7) = 16$

e $2(-3g + 7) = -16$

g $-3(-2h - 5) = 33$

i $4(5x - 1) = -24$

k $2(7 + 2x) = 38$

m $5(2x - 6) = 100$

o $9(8 - x) = 45$

q $20 = 2(3x + 11)$

s $18 = 3(2x + 14)$

u $26 = 13(8 - 4x)$

b $3(2x + 1) = 15$

d $-5(3x + 8) = -10$

f $-2(3x - 5) = 28$

h $-3(-2x + 7) = -45$

j $-4(5x + 6) = -104$

l $-3(4 - x) = -12$

n $6(3x + 15) = -72$

p $10(7 - 3x) = 20$

r $16 = 4(2x + 9)$

t $-5 = 5(3x + 1)$

v $32 = 8(16 - 5x)$

2 Ryan attempted to solve the equation $9(4x - 6) = 18$. His working is shown below:

$$9(4x - 6) = 18$$

$$4x - 6 = 9$$

$$4x = 15$$

$$x = \frac{15}{4}$$

a What was his mistake?

b Solve the equation correctly.

3 Solve the following equations:

a $3(w + 8) + 5 = 44$

c $-5(g + 4) + 9 = -41$

e $6(q + 1) - 5 = -23$

g $-8(x + 4) - 3 = -59$

i $2(x - 9) - 4 = 16$

b $9(x - 7) + 8 = 26$

d $-7(x + 5) + 4 = -17$

f $4(x + 6) - 8 = 48$

h $-2(x - 2) - 7 = -29$

j $5(x - 3) - 7 = -52$

4 Solve the following equations:

a $-\frac{y}{3} + 10 = 17$

c $\frac{5x}{8} - 9 = -4$

e $\frac{5x}{8} + 11 = 21$

b $-\frac{u}{4} + 15 = 8$

d $\frac{8c}{3} + 5 = -11$

f $\frac{-3c}{4} + 5 = 14$

$$\text{g} \quad -\frac{3x}{4} + 5 = -7$$

$$\text{i} \quad \frac{4x}{7} - 6 = -14$$



$$\text{h} \quad \frac{3x}{4} - 5 = -11$$

$$\text{j} \quad -\frac{5x}{6} - 6 = 14$$

5 Solve the following equations:

$$\text{a} \quad \frac{x-9}{5} + 4 = 7$$

$$\text{c} \quad \frac{x+2}{3} - 25 = -22$$

$$\text{e} \quad \frac{2x-12}{3} = 0$$

$$\text{g} \quad \frac{8x+4}{5} = -4$$

$$\text{i} \quad \frac{-13-4r}{3} = -15$$

$$\text{b} \quad \frac{x-3}{5} - 8 = -7$$

$$\text{d} \quad \frac{x+16}{3} + 5 = 3$$

$$\text{f} \quad \frac{3x+6}{2} = 15$$

$$\text{h} \quad \frac{-3t-6}{7} = -3$$

$$\text{j} \quad \frac{-9+5x}{2} = -22$$

6 Solve the following equations:

$$\text{a} \quad \frac{2(x-4)}{3} = 8$$

$$\text{c} \quad \frac{-3(x-3)}{4} = 0$$

$$\text{e} \quad \frac{9(x+10)}{2} = -6$$

$$\text{g} \quad \frac{7(x-9)}{5} = 14$$

$$\text{b} \quad \frac{5(x+1)}{2} = 10$$

$$\text{d} \quad \frac{6(x+5)}{7} = 12$$

$$\text{f} \quad \frac{4(x-2)}{3} = -3$$

$$\text{h} \quad \frac{2(x+11)}{9} = 8$$

7 Solve the following equations:

$$\text{a} \quad 2\left(\frac{x}{3} + 1\right) = 6$$

$$\text{c} \quad 4\left(\frac{x}{5} - 3\right) = -12$$

$$\text{e} \quad 18 = 2\left(\frac{x}{5} - 7\right)$$

$$\text{g} \quad 8\left(4 + \frac{x}{2}\right) = 48$$

$$\text{b} \quad 5\left(\frac{x}{2} + 7\right) = 60$$

$$\text{d} \quad 3\left(\frac{x}{4} - 10\right) = 15$$

$$\text{f} \quad 40 = 10\left(\frac{x}{6} - 1\right)$$

$$\text{h} \quad -4\left(12 - \frac{x}{3}\right) = -64$$